Manish

B.Tech., Chemical Engineering
Minor in AI and Data Science
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Examination	University	Institute	Year	CPI / %
Graduation (B.Tech.)	IIT Bombay	Indian Institute of Technology Bombay	2027	**9.74**
Intermediate	CBSE	Kendriya Vidyalaya No. 2, Agra	2023	96.00%
Matriculation	CBSE	Kendriya Vidyalaya No. 2, Agra	2021	99.00%

RESEARCH INTERESTS

Numerical **optimisation** and **machine learning** for atomistic simulation: transition-state and minimum-energy-path search, machine-learned interatomic potentials (MLIPs), Hessian-based preconditioning, scientific machine learning.

SCHOLASTIC ACHIEVEMENTS

- Awarded the **Institute Academic Prize**, IIT Bombay, for exceptional academic performance [2025]
- Secured **1st position** among **64,000+** participants in the **Google Solution Challenge** [2025]
- Received the **AP grade** (top ∼1% of class) in **3 core** chemical engineering courses [2024-25]
- Achieved a perfect **10.0/10.0 SPI** in the third semester [2024]

PUBLICATIONS

Preprints *equal contribution

[1] **Manish*** and A. Dayama*, "Leveraging spatial-temporal graph neural networks for multi-store sales forecasting", *arXiv:2511.19267* (2025).

[2] **Manish***, A. Pandey* and S. Das*, "Questions that determine the 'babu': a comprehensive analysis of India's UPSC Civil Services Mains Examination General Studies questions", *SSRN 5077877* (2024).

In preparation

[3] **Manish** and S. Vijay, "Machine-learned Hessian preconditioning for nudged elastic band transition-state searches", *arXiv preprint planned for late 2026.*

[4] **Manish** and S. Choudhary, "The hot-season peak in maternal and neonatal deaths in India: a national district-level analysis, 2009–24".

RESEARCH EXPERIENCE

ML-Hessian Preconditioning for NEB | B.Tech. Project [May 2025 – Present]

Guide: Prof. Sudarshan Vijay, Department of Chemical Engineering, IIT Bombay

- Finding a reaction's **transition state** with the nudged elastic band (NEB) takes hundreds of expensive **DFT** force calls; cutting that cost by preconditioning the optimiser with **ML-predicted curvature**
- Combined a projected **MACE-MP-0 Hessian** (perpendicular block) with an analytical **spring-force Jacobian** (parallel block) inside a safeguarded **L-BFGS** optimiser
- Diagnosed the breakdown of Hessian preconditioning along a band (**indefinite curvature** near the saddle, **subspace leakage**) and fixed it with a projected, curvature-floored operator; solved the tridiagonal spring-force Jacobian directly
- Benchmarked on **60 systems** (OC20NEB, Transition1x, diffusion) under a pre-fixed protocol with a sealed held-out set: converged on **33/60 vs. 28/60** for FIRE and **22/31 vs. 14/31** on held-out systems
- Needed a median **0.18**× FIRE's band evaluations where both converged (95% CI 0.14–0.23, $n = 24$; excludes Hessian construction), with paths agreeing to 0.005 Å per image
- Validating the preconditioner with **DFT (VASP)** forces on 10–15 surface-catalysis and diffusion systems

Hot-Season Peak in Maternal and Neonatal Deaths | R&D Project **[2026 – Present]**
Guide: Prof. Saket Choudhary, Koita Centre for Digital Health, IIT Bombay

- Built a national **district-month panel** (772 districts, 2009–24) from HMIS records linked to **ERA5** climate and **NFHS-5** survey data; Poisson models with **district-by-year fixed effects**
- Found deaths per birth peak in **May–July**, the hottest months: **+42%** maternal and **+22%** neonatal vs. February–March (∼1,100 and 4,500 excess deaths a year), in 96% and 89% of districts
- Ruled out reporting artefacts with **5 birth registers** as denominators; confirmed the neonatal peak in NFHS-5 (RR 1.21 vs. 1.23); temperature reproduces **69%** of the maternal seasonal pattern
- Showed the excess is largest in states with **weaker care around birth**; extending the framework to climate drivers of malaria and dengue

UPSC Civil Services Examination Analysis | SURP **[Jul 2024 – Sep 2024]**
Guide: Prof. Syaamantak Das, Summer Undergraduate Research Programme, IIT Bombay | Preprint [2]

- Applied **BERTopic** topic modelling to **1,000+** General Studies questions over a decade (2013–23) to track topic evolution and test for subject-level bias
- Identified a sustained rise in current-affairs, governance, environment and technology questions

KEY PROJECTS

Quant Auditor: Fine-Tuned LLM Agent | IE624 Course Project **[Oct 2026 – Present]**
Generative and Agentic AI, IIT Bombay

- Designed a controlled benchmark of **2,921 matched pairs** of flawed and clean trading backtests (13 strategy templates, 8 injected flaw types, 20 years of real prices) with a sandboxed multi-turn tool environment
- Fine-tuned **Qwen3-4B** with **QLoRA SFT** so every verdict cites evidence (57% → 100% of answers); diagnosed a label-leaking teacher as the main error source before moving to **GRPO**

STGNN for Multi-Store Sales Forecasting | EE782 Course Project **[Aug 2025 – Nov 2025]**
Guide: Prof. Amit Sethi, Department of Electrical Engineering | Preprint [1]

- Built a **spatiotemporal GNN** (learned adjacency, dilated temporal convolutions, graph convolution) forecasting 45 stores jointly; lowest overall error vs. ARIMA, LSTM and XGBoost
- Reached **2.08% MAPE** on Store 37, nearly halving ARIMA's error; the learned graph recovered store clusters without geographic metadata

GenAI for Drug Discovery | Google Solution Challenge **[Mar 2025 – Jul 2025]**
Secured 1st position among 64,000+ participants (team project)

- Trained a protein-conditioned **RNN** molecule generator with a BiLSTM-attention protein encoder on **550K+** BindingDB samples: 100% validity, 99.7% uniqueness
- Ranked candidates on a multi-objective score (QED, Lipinski, toxicity, solubility, binding) using **DiffDock** and **MolFormer**

Semantic Segmentation with DeepLabV3+ | Seasons of Code **[May 2025 – Jul 2025]**
Web and Coding Club, IIT Bombay

- Implemented **atrous convolutions** and **ASPP** from scratch in PyTorch: **82.5% mIoU**, 92.7% pixel accuracy, with a further 4.1% mIoU gain from augmentation and class balancing

PROFESSIONAL EXPERIENCE

FX Structuring Summer Associate | Morgan Stanley **[May 2026 – Jul 2026]**
Asia FX Structuring desk, Mumbai

- Built a parameterised pricing engine for **path-dependent FX** products (6 variants, 18 payoffs, 3 settlement conventions); validated on **40+** benchmark trades, all within **1 bp**
- Built an FX volatility screener (IV, RV, risk reversals, butterflies, carry) across G10 and EM Asia, and a correlation-network dashboard to identify currency clusters

Quant Developer | iWealth India LLP **[Aug 2025 – Sep 2025]**
Investment management firm (remote)

- Built an earnings-call pipeline (**Whisper** transcription, company matching, Slack summaries) and a filings summariser saving the team **5+ hours** of reading per day

TEACHING EXPERIENCE

Teaching Assistant | MA110, IIT Bombay **[Jan 2025 – May 2025]**
Linear Algebra and Differential Equations, Department of Mathematics

- Conducted weekly tutorials, doubt sessions and grading for **40+** first-year students

EXTRACURRICULAR ACTIVITIES

Mentorship

- Mentored **7 students** for 6 weeks in **Seasons of Code 2025** on vision-to-text: LSTM fundamentals, image-captioning pipelines and model fine-tuning
- Supervised **8 students** on fundamentals of **machine learning** and **NLP** in **Summer of Science 2025**
- Guided **10+ students** in building an NLP-based movie recommendation system at **WiDS'24**

Competitions

- Secured **1st** of **64 teams** in **ChemEBlitz**, Inter-IIT Chemical Engineering event
- Selected among the **top 30 teams** from IIT Bombay in **Smart India Hackathon 2023**

Volunteering

- Completed **50+ hours** of volunteer service under the **Dhruva Initiative, NSS**, mentoring grade 11–12 students in physics and chemistry for **JEE Main** preparation

RELEVANT COURSEWORK

Mathematics	Calculus | Linear Algebra and Differential Equations | Numerical Analysis | Computational Methods Lab | Online Learning and Optimisation
Machine Learning	AI and Data Science | Introduction to ML | Supervised Learning I & II | Advanced Topics in ML | Programming for Data Science
Chemical Engineering	Thermodynamics | Transport Phenomena | Heat and Mass Transfer | Process Fluid Mechanics | Reaction Engineering | Separation Processes | Process Control
Ongoing	Generative and Agentic AI | Introduction to Financial Engineering | Colloid and Interfacial Engineering

TECHNICAL SKILLS

Scientific Computing	Python, C++, NumPy, SciPy; quasi-Newton and preconditioned optimisation; HPC clusters
Atomistic Simulation	ASE, MACE, VASP
Machine Learning	PyTorch, Hugging Face, QLoRA fine-tuning, vLLM, scikit-learn
Tools	SQL, Git, Docker, FastAPI, LaTeX